

■ Git & GitHub Commands — Complete Cheat Sheet

Beginner-friendly reference for Git configuration, repositories, branches, history, undo, stash, tags, GitHub workflow, and everyday commands.

1. ■■ Git Configuration

Command	Explanation
git --version	Check Git version
git config --global user.name "YourName"	Set your Git username
git config --global user.email "you@example.com"	Set your Git email
git config --list	Show all Git configuration
git config user.name	Show current username
git config user.email	Show current email

2. ■ Create / Download Repository

Command	Explanation
git init	Initialize a new Git repository
git clone <url>	Clone/download a repository from GitHub
git status	Show current repository status

3. ■ Add & Commit

Command	Explanation
git add .	Stage all changes
git add <file>	Stage a specific file
git add -A	Stage all changes
git commit -m "message"	Save changes with a message
git commit -am "message"	Stage modified tracked files and commit

4. ■■ GitHub Push / Pull

Command	Explanation
git push	Push commits to remote repository
git push -u origin main	Set upstream and push
git pull	Fetch + merge changes from remote
git fetch	Download remote changes without merging

5. ■ Remote Repository

Command	Explanation
git remote -v	List remote repositories
git remote add origin <url>	Add GitHub remote
git remote remove origin	Remove remote
git remote rename origin upstream	Rename a remote
git remote get-url origin	Show remote URL

Command	Explanation
git remote set-url origin <url>	Change remote URL

6. ■ Branch Commands

Command	Explanation
git branch	List branches
git branch <name>	Create a new branch
git switch <name>	Switch to a branch
git switch -c <name>	Create and switch to a branch
git branch -d <name>	Delete a branch
git branch -D <name>	Force delete a branch
git branch -M main	Rename current branch to main

8. ■ Git History

Command	Explanation
git log	View complete commit history
git log --oneline	Compact commit history
git log --graph	Show branch history graph
git log --oneline --graph --all	Compact graph of all branches
git show <commit>	Show details of a commit
git diff	Show unstaged changes
git diff --staged	Show staged changes

9. ■■ Undo / Reset

Command	Explanation
git restore <file>	Discard changes in a file
git restore --staged <file>	Remove file from staging
git reset <file>	Unstage a file
git reset --soft HEAD~1	Undo last commit but keep changes
git reset --mixed HEAD~1	Undo commit and unstage changes
git reset --hard <commit>	Reset files to a specific commit
git revert <commit>	Create a new commit that reverses a previous commit

10. ■ Stash

Command	Explanation
git stash	Temporarily save changes
git stash pop	Restore latest saved changes
git stash list	List saved stashes
git stash apply	Apply a stash without deleting it
git stash drop	Delete a stash
git stash clear	Delete all stashes

11. ■■ Tags

Command	Explanation
git tag <tagname>	Create a tag
git tag	List tags
git tag -d <tagname>	Delete a tag
git push origin <tagname>	Push one tag to GitHub
git push origin --tags	Push all tags

13. ■ Search / Information Commands

Command	Explanation
git status	Check current status
git diff	See code changes
git log	See commit history
git show <commit>	See commit details
git branch	See branches
git remote -v	See remote repositories
git tag	See tags

14. ■ Help Commands

Command	Explanation
git help	General Git help
git help <command>	Open help for a command
git <command> --help	Quick command help
git <command> -h	Short command help

7. ■ Merge

Merges the **feature** branch into the current branch.

```
git switch main
git merge feature
```

Example branch idea: main → feature → Login / Navbar → merge → main

■ Useful Examples

Clone a repository:

```
git clone https://github.com/username/project.git
```

Basic workflow:

```
git status
git add .
git commit -m "Added new feature"
```

Push to GitHub:

```
git push -u origin main

After setting upstream:
git push
```

Remote setup:

```
git remote add origin https://github.com/username/project.git
```

Older branch command:

```
git checkout <name>
```

Modern alternative:
git switch <name>

■ Example Git History Output

```
git log --oneline
```

```
a82f31d Added login page  
91bc421 Updated navbar  
45ca782 Initial commit
```

■ ■ Be careful with: **git reset --hard**
It can permanently remove uncommitted work.

■ Stash Example

```
git stash  
git switch main
```

Later:
git stash pop

■ ■ Tag Example

```
git tag v1.0.0  
git push origin v1.0.0
```

Tags are commonly used for project versions/releases.

12. ■ .gitignore

Create a file named **.gitignore**.

```
node_modules/  
.env  
*.log  
dist/
```

Explanation: Tells Git which files/folders should not be tracked.

■ Most Important Commands for Beginners

#	Command
1	git --version
2	git config
3	git init
4	git clone
5	git status
6	git add .
7	git commit -m "message"
8	git push
9	git pull
10	git fetch
11	git branch
12	git switch
13	git merge
14	git remote -v
15	git log
16	git diff
17	git restore
18	git stash
19	git tag
20	git help

■ Real Project Workflow

For portfolio / Java / frontend projects:

```
# 1. Create repository git init # 2. Check files git status # 3. Add files git add . # 4. Commit git commit -m "Initial commit" # 5. Connect GitHub git remote add origin # 6. Set main branch git branch -M main # 7. Upload git push -u origin main
```

■ When you make changes later:

```
git status git add . git commit -m "Updated portfolio" git push
```

■ Remember This Flow

YOUR CODE	↓
git add .	↓
STAGING AREA	↓
git commit -m "..."	↓
LOCAL REPOSITORY	↓
git push	↓
GITHUB ■	

Quick reminder: Git manages version history locally; GitHub hosts and shares remote repositories.